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The Evidence of Climate Change Found In Our Weather

Introduction

We all know from books, conversations, and reading the news that there is irrefutable evidence climate change is real. But how drastic is this proof? Is evidence of climate change extreme enough that we can see it even in basic weather statistics? And if so, how large are the effects of climate change on our weather?

Data

I used a number of datasets from the National Center for Environmental Information. First, I combined the Daily Summaries - which include the daily maximum temperature, minimum temperature and precipitation - for the 20 largest cities in the US from 2000 to 2024. Second, I combined the Storm Events Database's reports of all tornadoes that occurred in the US from 2006 to 2024.

Methodology and Results

I first combined all of my Daily Summary datasets into one large dataset and carried out some data cleaning steps, including dropping observations with NAs and creating columns for the month and year. After this I created a few columns to use in my regressions. First, I created a daily range column representing the difference between the maximum and minimum temperature. I also created three variance columns: yearly variance of the maximum temperature, yearly variance of the minimum temperature, and yearly variance of the precipitation. In addition, I created a daily spread column that represented the difference between the maximum and minimum temperature each day. The last three columns I created were dummy variables representing whether there was an extreme temperature, an extreme high temperature, or an extreme low temperature. I created these by taking the 5th percentile of the minimum temperature and 95th percentiles of the maximum temperature in each city in 1999. If a maximum temperature in any city in the year 2000 or onwards was above that city's maximum cutoff value, the temperature was classified as an extreme max. If the minimum temperature was below that city's minimum cutoff value, it was classified as an extreme minimum. To be classified as an extreme temperature, it had to be either an extreme max, min, or both.

The second dataset I used included every tornado episode reported from 2006 to 2024 in the US. The NCEI would only allow 500 tornado episodes to be downloaded at a time. So, I downloaded approximately 5 csvs per year for every year 2006 to 2024. Once these were downloaded, I began by combining all these individual datasets into one dataset, which contained all the tornado episodes reported in all US states 2006 - 2024; I also performed some basic data cleaning steps such as dropping unnecessary columns and creating month and year columns. On a given day, one tornado was often reported within a state as multiple different tornado episodes. However, there was no obvious standardized method of determining how many episodes each single tornado was considered to be. Some tornadoes were reported as one episode per state per day and other tornadoes were reported as dozens of episodes in the same state on the same day. So, I choose to base my regression on the measure of whether or not a state had at least one episode of a tornado in a given day to prevent reporting methods biasing the tornado count measures. So, I also added two columns to this dataset: a dummy for whether or not each specific state had a tornado on that date and the total number of states with at least one tornado on that date. Next, I created an empty dataframe with an observation representing every state every day 2006-2024. While my original data frame only had data points for days that had tornadoes, this allowed me to also have data points for every state each day regardless of whether it had a tornado or not. I combined my original tornado dataset with the new empty dataset of all days and states, setting the dummy for whether or not a state experienced a tornado to 0 for all the state-day observations not represented in the original tornado database.

As I ran 15 tests, under the null the expected number of false positives using a 5% significance level would be .75 (.05*15). So, I chose to apply Bonferroni corrections to the p values so that under the null there is a 5% chance of any one of these falsely rejecting instead. In order to consider one of the tests significant with this correction, the p value had to be less than .0033 (.05/15).

Predicting Changes in Temperature and Precipitation

<u>Extreme Values</u>

The first regression I ran predicted whether extreme temperatures are growing more common due to climate change. I then ran three logistic regressions on the values of the extreme temperature columns from the year 2000 - 2024 using a time trend, city fixed effect, and controlling for the month in order to predict the log odds of extreme high temperatures, extreme low temperatures, and general extreme temperatures. The coefficient on the time trend for the regression on extreme high temperatures was .029 with a p value of approximately 0 (reported by R as “< 2e^-16”), suggesting extreme high temperatures are becoming significantly more frequent. The coefficient on the time trend for the regression on extreme low temperatures was -.038 with a p value of approximately 0, suggesting extreme low temperatures are becoming significantly less frequent. However, when looking at extreme temperatures on the whole, the p value on the coefficient for the time trend was not significant.

I also found evidence of temperatures increasing over time (discussed in the next section). This suggests that while we are facing increasing rates of extreme heat, using historical standards, this may actually be due more to all temperatures increasing rather than an increasing likelihood of extreme temperatures. Given I used data from 1999 to determine the cutoff values that are used to classify extreme temperatures, the increase in extreme high temperatures and decrease in extreme low temperatures could be due to future years being hotter than 1999.

<u>Temperature and Precipitation Changes</u>

As mentioned in the previous paragraph, I also ran a linear regression to predict the maximum temperature each day using a time trend with city fixed effects and a control for the month. This produced a coefficient on the time trend of .09 with an approximately 0 p value. I also ran this to predict the minimum temperature each day, finding a coefficient of .0839 on the time trend with an approximately 0 p value. These findings suggest that both maximum and minimum daily temperatures are increasing over time. Finally, I ran the same regression to predict precipitation levels. The coefficient on the time trend was .000306, suggesting that precipitation is also increasing over time. The p value of .002 on this coefficient was still significant with the Bonferroni correction.

Trying an Alternative Model

I also considered that the climate might be changing at an ever increasing rate rather than a linear rate. To test this, I tried using a model with an additional variable representing the squared time trend. Comparing the R-squared values, the original regression predicting the maximum temperature discussed in the last paragraph had an R-squared of .7134 while the same regression with this additional squared time trend term had an R-squared of .7077. So using the R-squared as a metric for the fit, I choose to stick to my original assumption that the maximum temperature is changing at a linear rate. However, the R-squared for the original regression predicting the minimum temperature (.7375) was very slightly lower than that of the regression with the additional squared term (.7377). So, an additional squared time series term here may slightly improve the model. The R-squared for the regression predicting precipitation also slightly decreased from .0167 to .0164 with this additional term, so the squared time series did not seem to improve this model.

Evaluating the results of the model for predicting the minimum temperature (the only one where adding the squared time trend improved the R-squared), it appears that the average minimum temperature is increasing at least .058 each year. The positive coefficient on the squared time trend suggests that this process is also speeding up. Both the p values on the squared time trend coefficient (.0009) and time trend coefficient (approximately 0) were significant with a Bonferroni correction.

Variance and Daily Spread of Temperature and Precipitation Over Time

The next set of regressions I ran were on the daily spread and yearly variance. The regression predicting the daily spread using a time trend, city fixed effect, and controls for the month had a coefficient on the time trend of approximately .005 but the p value of this coefficient was .08. This was not significant at a 5% level even without a Bonferroni correction. So this offers no evidence that the typical daily range of temperatures is changing over time.

The regressions predicting the yearly variance of the maximum temperature with a time trend, city fixed effect, and controls for the month had a coefficient of approximately .08 and nearly 0 p value. The coefficients on the yearly variance of the minimum temperature and precipitation had coefficients of .2 and .0005 respectively. This suggests that weather is generally becoming more variable.

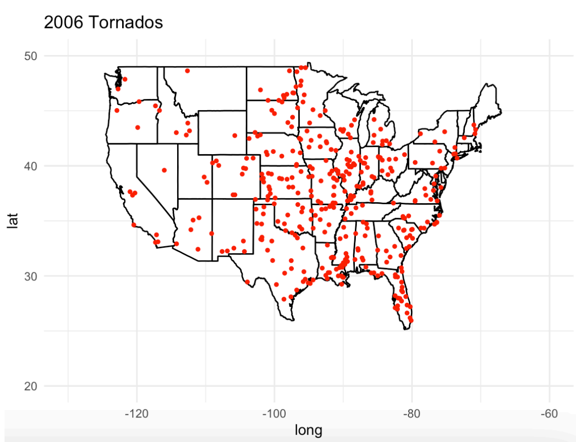
Predicting Changes in Tornado Frequency

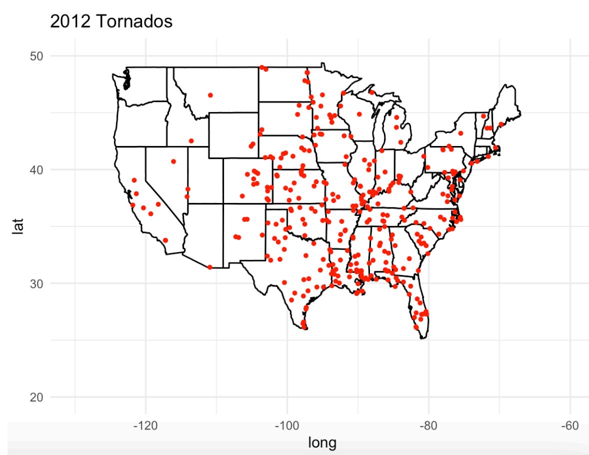
Tornado Frequencies Over Time

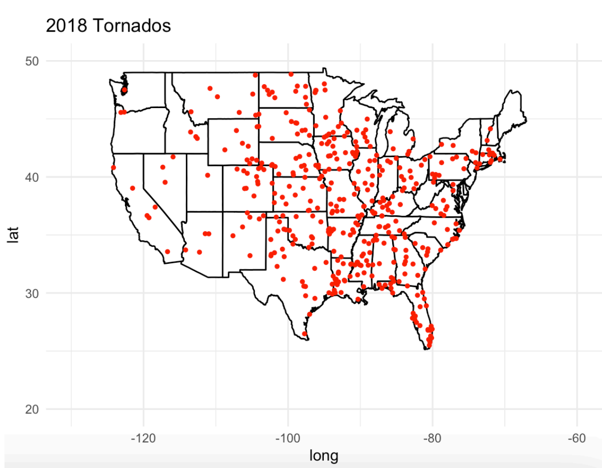
I ran one logistic and one linear regression with the tornado dataset. I first ran a logistic regression predicting the log odds that there was a tornado on a given day with a time trend, controls for the month, and state fixed effects. The p value on the time trend was not significant (.36) even without the Bonferroni correction. I also ran a linear regression predicting the total number of states with tornadoes on a given day. The coefficient on the time trend was .003 with a significant p value (.00003). So, from this data, there is no significant evidence that the log likelihood of a state having at least one tornado is going up over time but there is evidence that the number of states experiencing tornadoes is slightly increasing.

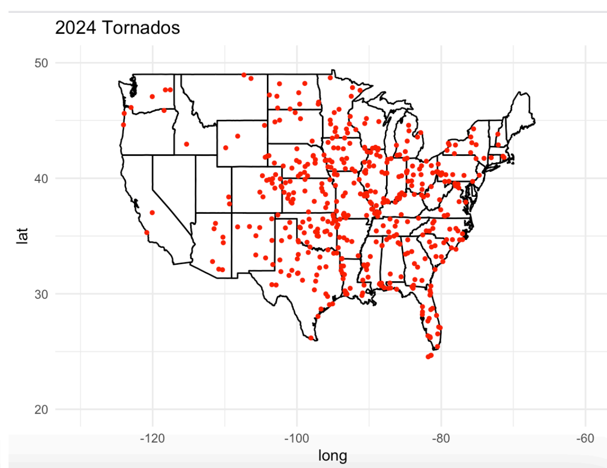
Tornado Locations

I finally looked into where the tornadoes were occurring to see if there were any pattern shifts in tornado locations. I used ggplot to graph all the tornadoes that occurred in the years 2006, 2012, 2018, and 2024 (pictured below). However, there were no clear trends in tornado locations across these four time snapshots.



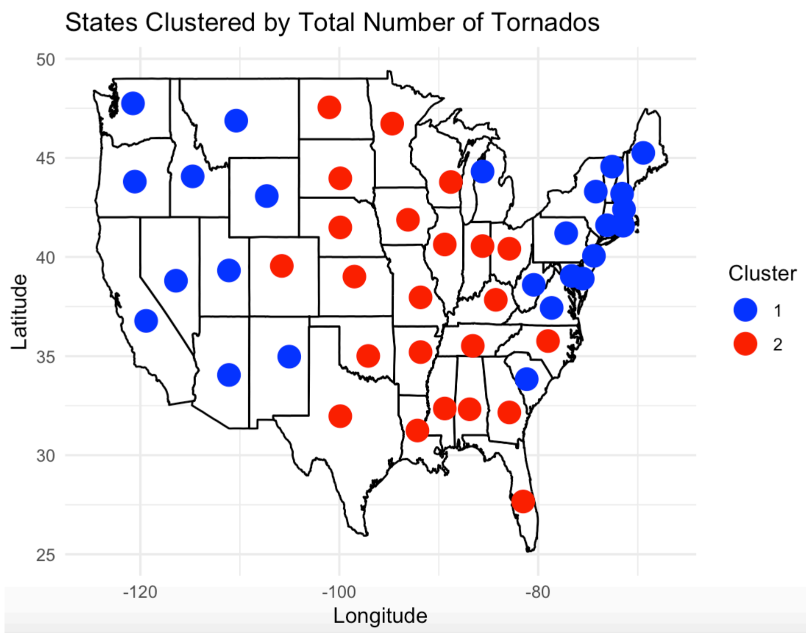




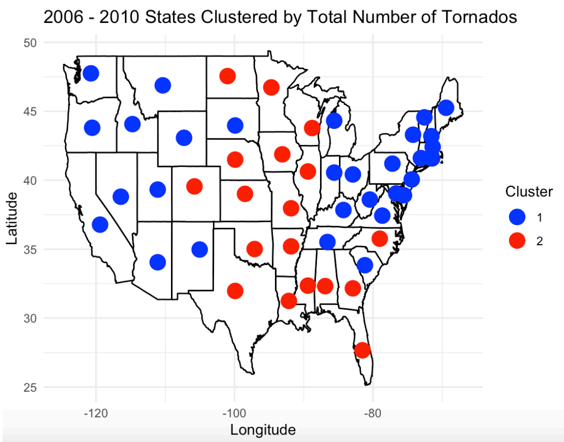


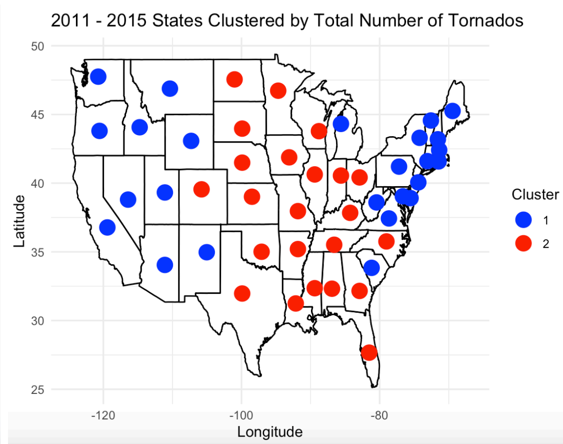
Clustering into High-Risk and Low-Risk Areas

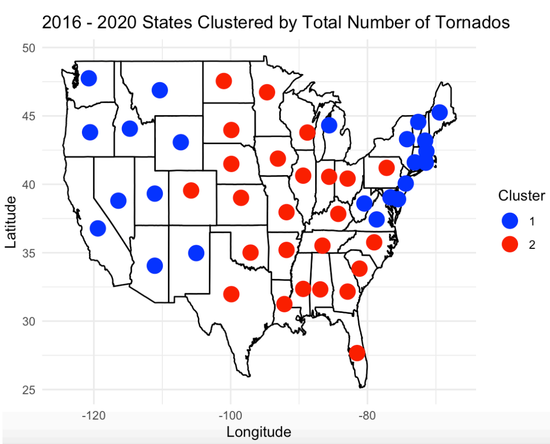
As an extension after we learned about K-means clustering, I also used the data on the total number of tornadoes in each state to do a very basic form of clustering. I created a variable that was the percent of the time each state had at least one tornado. Using this column, I used k-means clustering to split the data up into two clusters that represent higher tornado risk states and lower tornado risk states. The original dataframe only has longitude and latitude values for states with tornado episodes, so I combined this with a data frame that contained the approximate longitude and latitude of the middle of each state. Using these longitude and latitude values, I plotted a data point representing each state with the color of the point representing what cluster the state was in. The west half of the country and the upper half of the east coast appear to be at a lowest tornado risk.

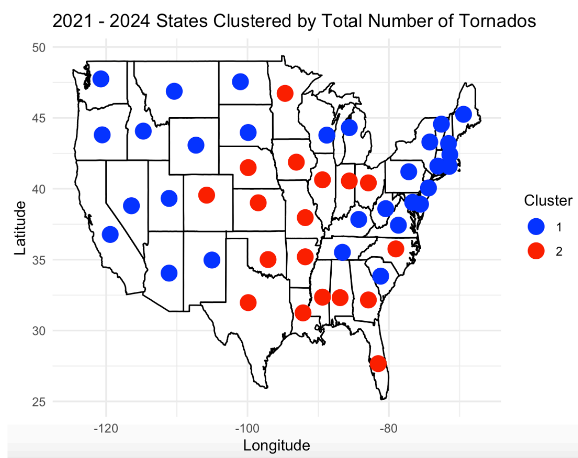


Focusing on tornado pattern changes, I also separated the data into four periods: 2006-2010, 2011-2015, 2016-2020, and 2021-2024 and re-did this process for each of those. Similar to the last section, there were no clear consistent patterns in which areas of the country changed color across time, so I find no strong evidence of climate change through shifts in where tornadoes are occuring.









So, the only evidence of climate change from patterns in tornadoes within my dataset is a slight increase in the average numbers of states with tornadoes each day. However, there is strong evidence of climate change using temperature and precipitation. Temperatures have been significantly increasing over time. There are also rising instances of extremely high temperatures and decreasing instances of extreme low temperatures. Particularly worryingly, there is some evidence in this dataset that temperatures may also be increasing at an increasing rate. Rainfall has also been increasing and becoming more variable. So, multiple measures of our basic daily weather offer strong evidence of climate change.